22 de diciembre de 2014

1) $h[n] = \left(\frac{1}{2}\right)^n u[n] + \left(\frac{1}{3}\right)^n u[n]$

a) $H[\Omega] = \frac{1}{1-\frac{1}{2}e^{j\Omega}} + \frac{1}{1-\frac{1}{3}e^{j\Omega}}$

$H[\Omega] = \frac{1-\frac{1}{3}e^{j\Omega} + 1-\frac{1}{2}e^{j\Omega}}{1-\frac{1}{3}e^{j\Omega}-\frac{1}{2}e^{j\Omega}+\frac{1}{6}e^{-2j\Omega}}$

Además $H[\Omega] = \frac{Y[\Omega]}{X[\Omega]}$

$Y[\Omega]\left(1-\frac{1}{3}e^{j\Omega}-\frac{1}{2}e^{j\Omega}+\frac{1}{6}e^{-2j\Omega}\right) = X[\Omega]\left(2-\frac{1}{3}e^{j\Omega}-\frac{1}{2}e^{j\Omega}\right)$

$y[n] - \frac{1}{3}y[n-1] - \frac{1}{2}y[n-1] + \frac{1}{6}y[n-2] = 2x[n] - \frac{1}{3}x[n-1] - \frac{1}{2}x[n-1]$

$y[n] - \frac{5}{6}y[n-1] + \frac{1}{6}y[n-2] = 2x[n] - \frac{5}{6}x[n-1]$

b) Dadas las especificaciones, el orden del filtro FIR debe ser 1000 con una transferencia:

$$\tilde{h}[n] = \begin{cases} h[n] & n \in [0, 999] \\ 0 & e.o.c \end{cases}$$

c) $y[n] = \sum_{k=0}^{1000} h(k)\,x(n-k)$ ← La convolución de los primeros 1000 puntos.

---

$$y[n] = (h * k)[n] = \sum_{k=0}^{\infty}\left(\frac{1}{3}k + \frac{1}{2}k\right)x[n-k]$$

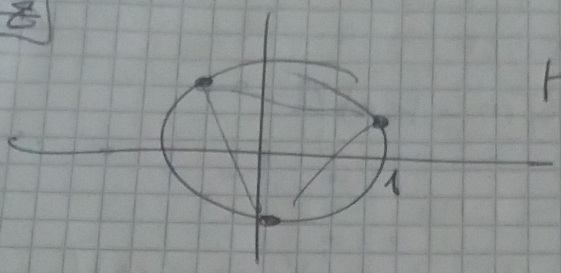
$$= 2x[n] + \sum_{k=1}^{\infty}\left(\frac{1}{3^k} + \frac{1}{2^k}\right)x[n-k]$$

$$= 2x[n] + \sum_{k=0}^{\infty}\left(\frac{1}{3^{k+1}} + \frac{1}{2^{k+1}}\right)x[n-1-k]$$

$$= 2x[n] + \left(\frac{1}{2} + \frac{1}{3}\right)y[n-1]$$

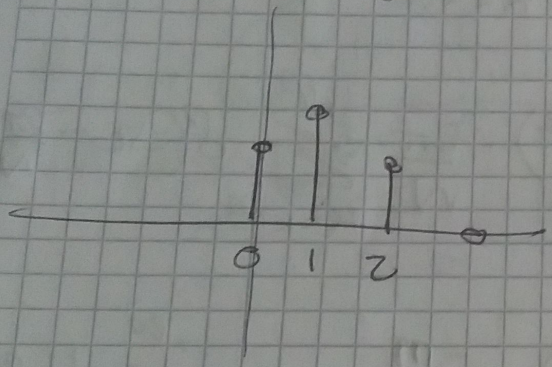
$$y[n] - \frac{5}{6}y[n-1] = 2x[n]$$

e)

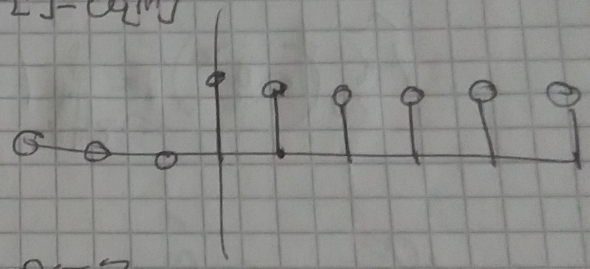


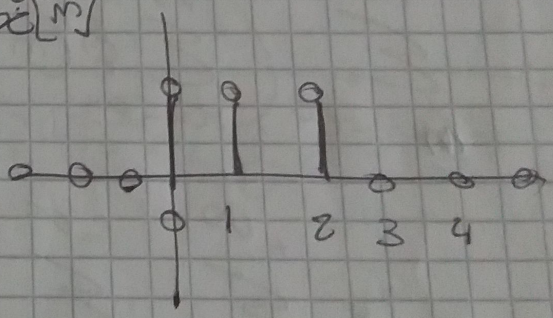

$$H[k] = H(e^{0(k\pi + \varphi_0)})$$

$$\tilde{x}[k] = \begin{cases} x[n] , & n = 0, 1, 2 \\ 0 , & \text{otro caso.} \end{cases}$$



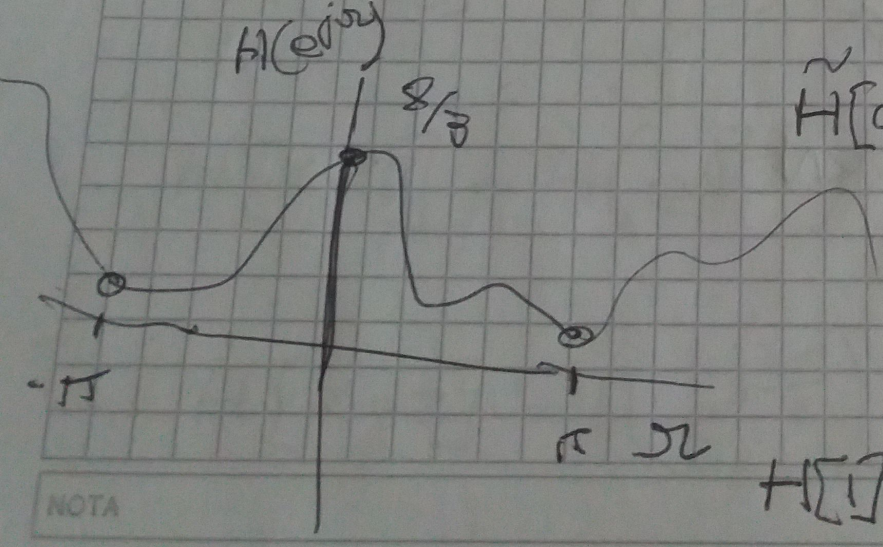

$x[n] = u[n]$






$\tilde{x}[n]$

$$H(e^{j\Omega}) = \frac{1}{1 - \frac{3}{4}e^{-j\Omega} + \frac{1}{8}e^{-2j\Omega}}$$




$$\tilde{H}[0] = H(e^{0}) = \frac{1}{1 - \frac{3}{4} + \frac{1}{8}}$$

$$H[0] = \frac{8}{3}$$

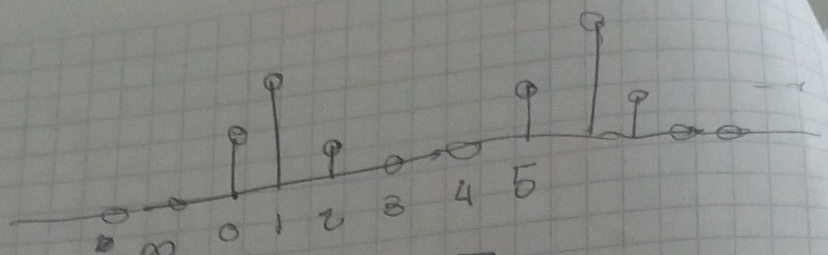
$$H[1] = H(e^{j\pi})$$

NOTA

Coloquio de Señales y Sistemas (66.74 y 86.

15 de Diciembre de 2014

Aquellos desarrollos
n válidos. Se pide
iento en todos los
s otros ejercicios
cuatrimestre




$$\tilde{a}_N[m] = \sum_{k=-\infty}^{+\infty} a[m-kN]$$

$$= \sum_{k=-\infty}^{+\infty} a[m] * \delta[m-kN]$$

$$= a[m] * \sum_{k=-\infty}^{+\infty} \delta[m-kN]$$

$$\tilde{H}_N(e^{j\Omega}) = \frac{1}{2\pi} H(e^{j\Omega}) \cdot \frac{2\pi}{N} \sum_{k=-\infty}^{+\infty} \delta\left(\Omega - \frac{2\pi}{N}k\right)$$

$$\sum_{m=-\infty}^{+\infty} \left( \sum_{k=-\infty}^{+\infty} \delta(m-kN) \right) z^{-m} = 2\pi \sum_{k=-\infty}^{+\infty} \frac{1}{N} \delta\left(\Omega - \frac{2\pi k}{N}\right)$$

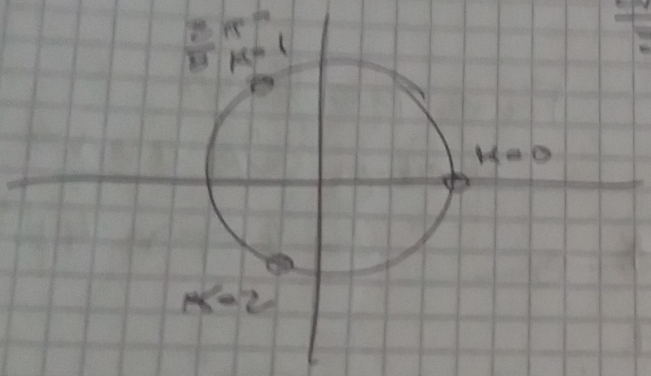
$$= \sum_{m=-\infty}^{+\infty} \sum_{k=-\infty}^{+\infty} z^{-kN} \delta(m-kN) = \frac{2\pi}{N} \sum_{k=-\infty}^{+\infty} \delta\left(\Omega - \frac{2\pi k}{N}\right)$$

$$= \sum_{k=-\infty}^{+\infty} z^{-kN}$$

$$\tilde{H}_N(e^{j\Omega}) = \frac{2\pi}{N} H(e^{j\Omega}) \sum_{k=-\infty}^{+\infty} \delta(\Omega - k\frac{2\pi}{N})$$

$$\boxed{\tilde{H}_N(e^{j\Omega}) = \frac{2\pi}{N} H(e^{j\frac{2k\pi}{N}})}$$

$H(z)$



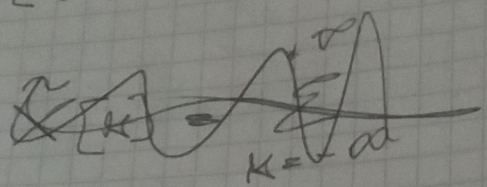

$$H(e^{j\frac{2k\pi}{3}}) = \frac{1}{1 - \frac{3}{4}e^{-j\frac{2k\pi}{3}} + \frac{1}{8}e^{-j\frac{4k\pi}{3}}}$$

$$H[k=0] = H(e^0) = \frac{1}{1 - \frac{3}{4} + \frac{1}{8}} = \boxed{\frac{8}{3}}$$

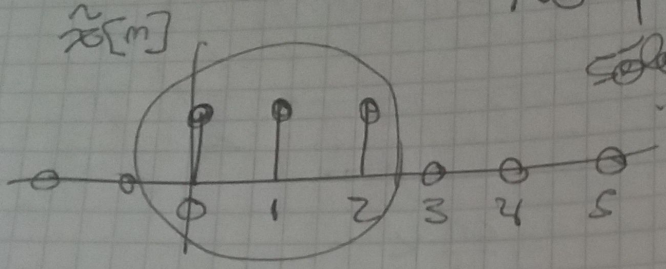
$$H[k=1] = H(e^{j\frac{2\pi}{3}}) = \boxed{\frac{4}{7} - \frac{4\sqrt{3}}{21}j}$$

$$H[k=2] = H(e^{j\frac{4\pi}{3}}) = \boxed{\frac{4}{7} + \frac{4\sqrt{3}}{21}j}$$

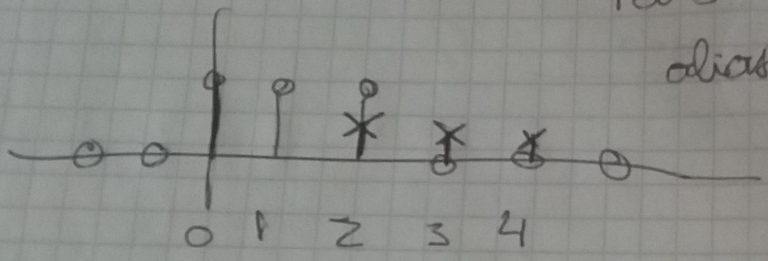
$\tilde{x}[n] = \delta(n) + \delta(n-1) + \delta(n-2)$



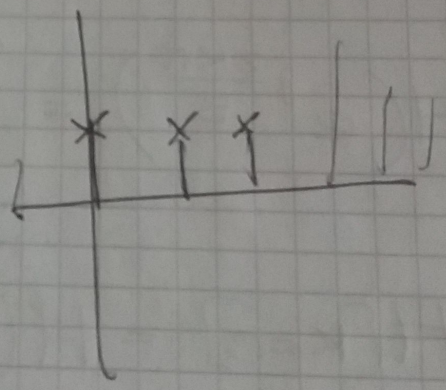

No puede agarrar solo 3 ptos...




Para no tener aliasing debe ser la DFT de 4 ptos.




$$H(z) = \frac{1}{1 - \frac{3}{4}z^{-1} + \frac{1}{8}z^{-2}}$$



$$= \frac{1}{\frac{1}{8}(z^{-1}-4)(z^{-1}-2)}$$

$$= \frac{8}{(4-z^{-1})(2-z^{-1})} = \frac{A}{4-z^{-1}} + \frac{B}{2-z^{-1}}$$

$$= \frac{2A - Az^{-1} + 4B - Bz^{-1}}{(4-z^{-1})(2-z^{-1})}$$

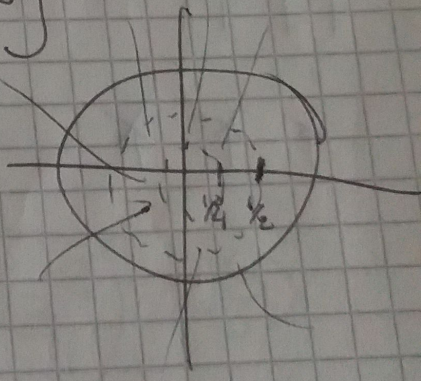
$$= \frac{(2A + 4B) - (A+B)z^{-1}}{(4 - z^{-1})(z - z^{-1})}$$

$$\begin{cases} 2A + 4B = 8 \rightarrow 2B = 8 \\ A + B = 0 \\ A = -B \end{cases} \qquad \begin{matrix} B = 4 \\ A = -4 \end{matrix}$$

$$H(z) = \frac{\frac{1}{2}}{\frac{1}{2}(z - z^{-1})} \cdot \frac{4}{} - \frac{4}{(4 - z^{-1})} \cdot \frac{\frac{1}{4}}{\frac{1}{4}}$$

$$= \frac{2}{1 - \frac{1}{2}z^{-1}} - \frac{1}{1 - \frac{1}{4}z^{-1}} \quad \begin{matrix} |z| > \frac{1}{2} \\ |z| > \frac{1}{4} \end{matrix}$$

$$= \mathcal{Z}\left[2 \cdot \left(\frac{1}{2}\right)^n u[n] - \left(\frac{1}{4}\right)^n u[n]\right]$$




Estable $\Rightarrow$ RoC contiene $e^{j\omega}$.

$\Rightarrow$ debe ser hacia afuera.

$$a[n] = \left(\frac{1}{2}\right)^{n-1} u[n] - \left(\frac{1}{4}\right)^n u[n]$$

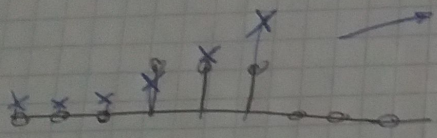
Aclaración: Todos los desarrollos deben estar debidamente justificados. Aquellos desarro
que no tengan una justificación clara en cada uno de sus pasos no serán válidos. Se
también prolijidad y letra clara. Para aprobar es necesario probar conocimiento en
ejercicios. Si un ejercicio no se aborda el examen no será aprobado aunque los otros
estén resueltos adecuadamente. Aclare en esta misma hoja nombre, padrón, co
práctica en que cursó la materia.

1. Sea un sistema cuya transferencia vale:

$a[n] = \left(\frac{1}{2}\right)^{n-1} u[n] - \left(\frac{1}{4}\right)^{n} u[n]$

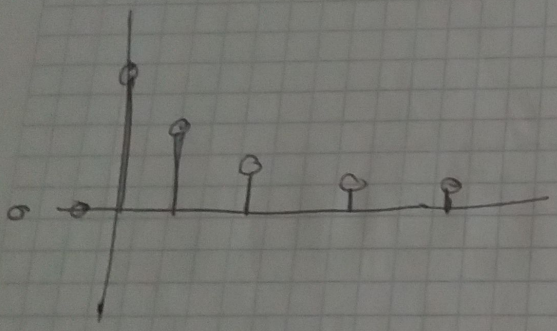
(es a derecha) y $h[n] = 0$, $n<0$.

$\Rightarrow$ causal.



~~Como mucho necesitaré~~

¿En los puntos depende de todos $a[n]$?



$$\mathcal{Z}[u[n]] = \frac{1}{1-z^{-1}}$$

$$\frac{8}{(z^{-1}-4)(z^{-1}-2)(1-z^{-1})}$$

$$= \frac{-8}{(z^{-1}-4)(z^{-1}-2)(z^{-1}-1)}$$

$$= -\frac{8}{3}\frac{1}{z^{-1}-1} + 4\cdot\frac{1}{z^{-1}-2} - \frac{4}{3}\frac{1}{z^{-1}-4}$$

$$= \frac{8}{3}\frac{1}{1-z^{-1}} - 4\frac{1}{2-z^{-1}} + \frac{4}{3}\frac{1}{4-z^{-1}}$$

$$= \frac{8}{3}\frac{1}{1-z^{-1}} - 4\,\frac{\frac{1}{2}}{\frac{1}{2}}\cdot\frac{1}{2-z^{-1}}$$

$$+ \frac{4}{3}\cdot\frac{\frac{1}{4}}{\frac{1}{4}}\,\frac{1}{4-z^{-1}}$$

$$= \frac{8}{3}\frac{1}{1-z^{-1}} - 2\,\frac{1}{1-\frac{1}{2}z^{-1}} + \frac{1}{3}\cdot\frac{1}{1-\frac{1}{4}z^{-1}}$$

↳

$$= \frac{8}{3}u[n] - 2\cdot\left(\frac{1}{2}\right)^n u[n] + \frac{1}{3}\cdot\left(\frac{1}{4}\right)^n u[n]$$

$$= \frac{8}{3}u[n] - \left(\frac{1}{2}\right)^{n-1} u[n] + \frac{1}{3}\left(\frac{1}{4}\right)^n u[n]$$

$$Y(z) = X(z)\left(\frac{8}{(z^{-1}-4)\,1}\right)$$

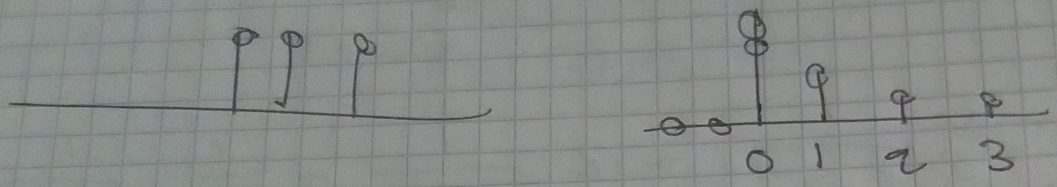
$$Y(z) = \left(1 - \frac{3}{4}z^{-1} + \frac{1}{8}z^{-2}\right)X(z)$$

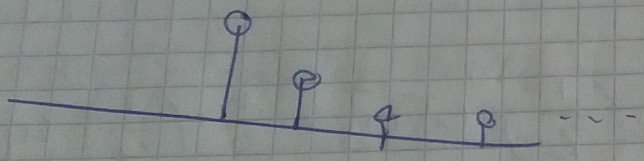
$$y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] = x[n]$$

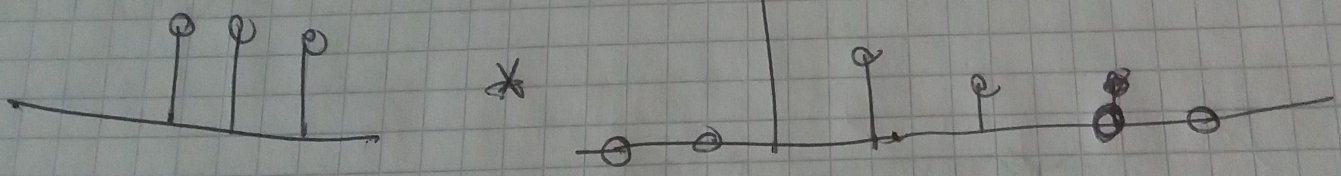
$$\boxed{y[n] = \frac{3}{4}y[n-1] - \frac{1}{8}y[n-2] + x[n]}$$

$$h[n] * (\delta(n) + \delta(n-1) + \delta(n-2))$$

$$= h[n] + h[n-1] + h[n-2]$$

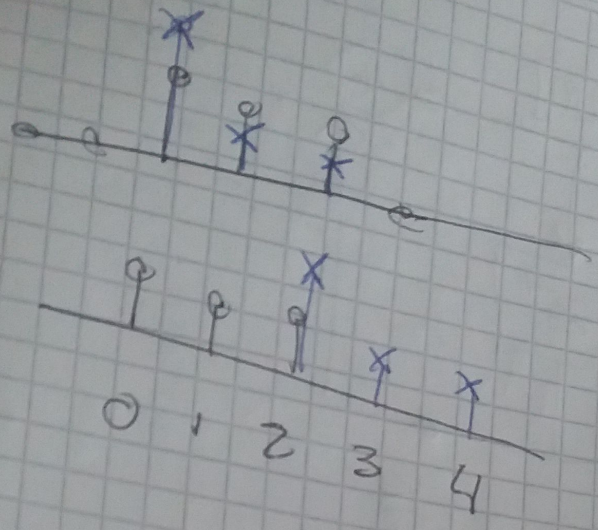




# $\tilde{h}[n] = \begin{cases} h[n], & n = 0,1,2 \\ 0, & \text{otro caso.} \end{cases}$

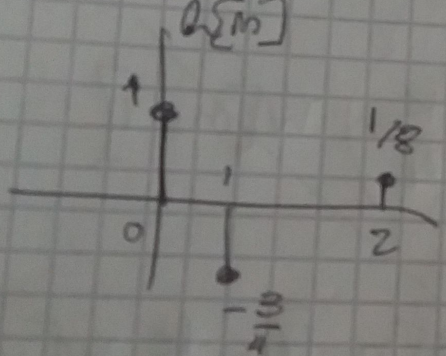
$$\tilde{x}[n] = \begin{cases} x[n], & n = 0,1,2. \\ 0, & \text{otro caso.} \end{cases}$$




$N = 5$ puntos.

$$H(z) = 1 - \frac{3}{4}z^{-1} + \frac{1}{8}z^{-2}$$

$$a[n] = 1 - \frac{3}{4}\delta(n-1) + \frac{1}{8}\delta(n-2)$$

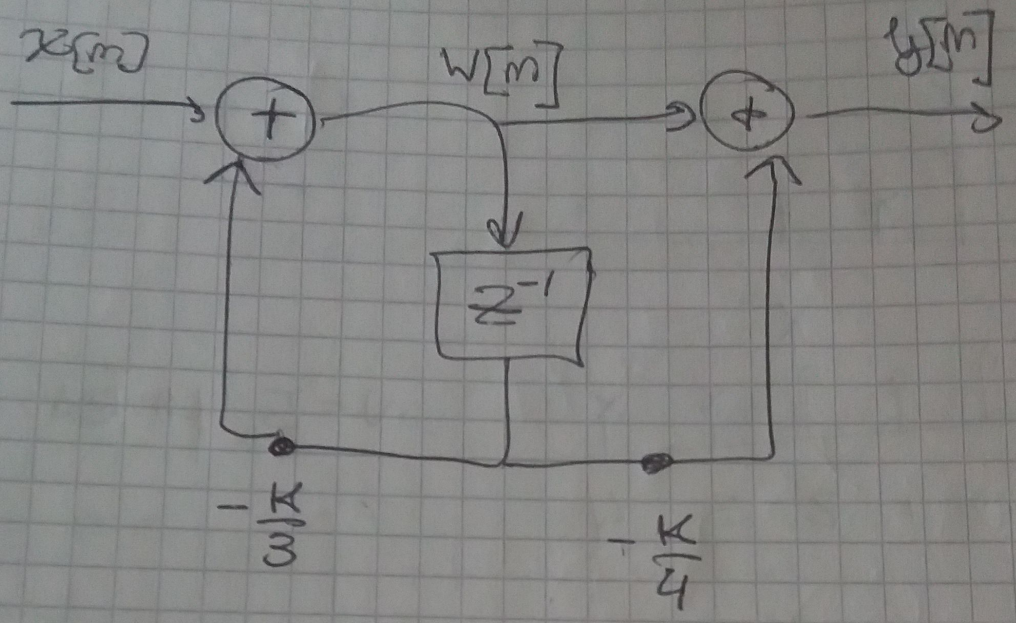



$$H(e^{j\omega}) = 1 - \frac{3}{4}e^{-j\omega} + \frac{1}{8}e^{-2j\omega}$$

$$H(e^{j\frac{2k\pi}{3}}) = 1 - \frac{3}{4}e^{-j\frac{2k\pi}{3}} + \frac{1}{8}e^{-j\frac{4k\pi}{3}}$$

Cuidado con estos ejes.

1o



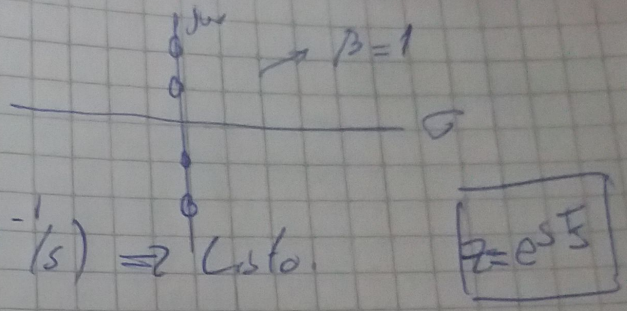

$$\begin{cases} w[n] = x[n] - \frac{K}{3} w[n-1] \\ y[n] = w[n] - \frac{K}{4} w[n-1] \end{cases}$$

$$y[n] = x[n] - \left(\frac{K}{3} + \frac{K}{4}\right) w[n-1]$$

$$= x[n] - \left(\frac{K}{3} + \frac{K}{4}\right)\left[x[n-1] - \frac{K}{3} w[n-2]\right.$$

El sistema que genera eco tiene transferencia

$H(s) = 1 + \beta e^{-s\tau}$ $\quad s \in \mathbb{C}$




Si el sist. "anti-eco" tiene $H_2(s) = H^{-1}(s) \Rightarrow$ Estable.

$z = e^{sT}$

$$\begin{cases} H(s)^{-1} = \dfrac{1}{1+\beta e^{-s\tau}} \\ s \in \mathbb{C} \end{cases} \Rightarrow y(t) + \beta y(t-\tau) = x(t)$$

Usando $z = e^{sT} \Rightarrow z^{-s\tau} = (z^{-1})^{\alpha}$ con $z = \alpha T$

Si $\alpha = 1$, es decir $z = \beta T$

$H(z) = \dfrac{1}{1+\beta z^{-1}}$ → Si el filtro reconstructor tiene ganancia entonces $g_2$ está (compensa ADC).

$$H(e^{j\Omega}) = \frac{1}{1+\beta e^{-j\Omega}}$$

La ec en dif es: $y[n] + \beta y[n-1] = x[n]$